

A Rigorous, Reproducible Analysis of CNN Architectures for Music Genre Classification

A Replication and Extension Study on the GTZAN Dataset

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Motivation & Objectives

Motivation: The Challenge of Reproducible Results in MGC

Key Message: The field is noisy. We bring clarity and rigor.

- Music Genre Classification (MGC) is fundamental in MIR.
- GTZAN is the de-facto benchmark, yet reported accuracies vary widely.
- 2023 paper (Patil et al.) claims **99.41%** accuracy – an extreme outlier.
- Raises methodological concerns (e.g., potential **data leakage**).

Our Goals:

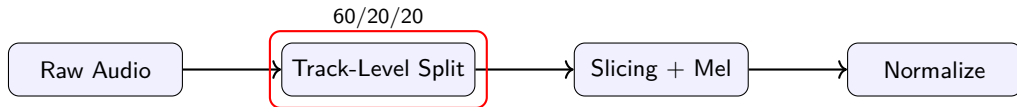
1. **Validate:** Re-implement the U-Net idea in a rigorous pipeline.
2. **Compare:** Benchmark against standard CNN architectures.
3. **Generalize:** Test robustness across diverse, unseen datasets.

Methodology

Methodology I: A Leak-Free Data Pipeline

Key Message: Reliability starts with disciplined data handling.

- Track-level train/val/test split (60/20/20) **before** any feature extraction.
- Prevents segments from same track leaking across splits.
- Standardization fit **only on training** set.
- Input: 128-bin Mel-spectrogram slices (time-frequency image proxy).



Methodology II: The Architectural Tournament

Key Message: Three representative CNN families compared under identical settings.

1. Efficient-VGG

- Lightweight VGG-style
- Strong baseline
- Fast inference

2. ResSE-AudioCNN

- Residual blocks
- Squeeze-Excitation
- Deeper representation

3. U-Net Audio Classifier

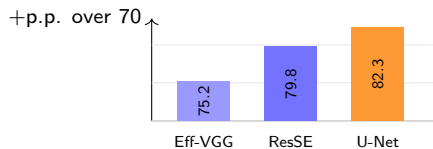
- Multi-scale encoder
- Skip connections
- Hypothesis: richer features

Core Results

Results on GTZAN: A Clear Performance Hierarchy

Key Message: Well-motivated complexity yields real gains.

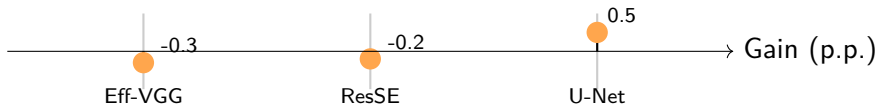
- Identical training protocol (SpecAugment)
- U-Net outperforms other CNN baselines
- Multi-scale feature fusion provides consistent gains



Ablation: Why Data Augmentation Matters

Key Message: Augmentation helps only when model capacity can exploit it.

- Full re-training without SpecAugment.
- U-Net: +0.5 p.p. gain with augmentation.
- Lower-capacity models: slight degradation (noise injection effect).
- Shows interaction between architecture depth and regularization.



Extended Analysis

Beyond Accuracy: Efficiency & Complexity

Key Message: U-Net balances accuracy and efficiency better than the deeper attention model.

Model	Test Acc.	Params (M)	Latency (ms)
Efficient-VGG	75.2%	0.03	14.8
ResSE-AudioCNN	79.8%	1.23	26.8
U-Net Audio Class.	82.3%	1.18	23.1

- Efficient-VGG: minimal but sacrifices accuracy.
- U-Net: competitive parameter count and faster than ResSE.
- Practical choice for deployment among high performers.

Cross-Dataset Generalization Strategy

Key Message: We test robustness under scale, cultural, and task shifts.

FMA Small

Open-domain scale

Larger, more diverse set.

exititTrain from scratch

Indian Music

Cultural shift

Timbre / scales differ.

exititFine-tune upper layers

Tabla Taala

Rhythmic task

Pattern emphasis.

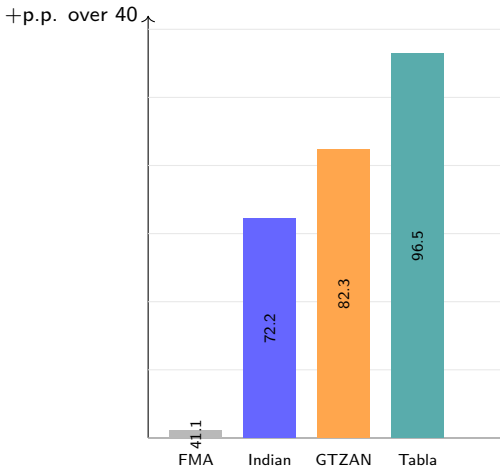
exititFine-tune classifier

Strategy: freeze early convolutional blocks for fine-tuning; adapt classifier head to domain/task.

Results: Robustness and Adaptation

Key Message: Strong transfer; open-domain diversity still challenging.

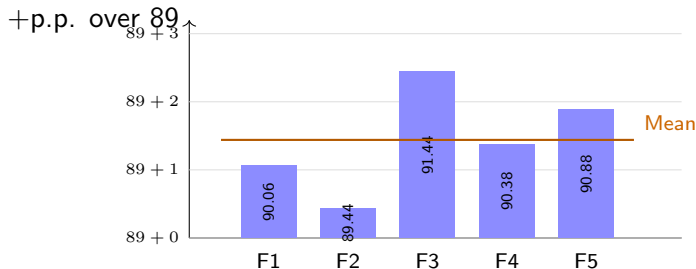
- Tabla Taala: **96.5%**
- GTZAN: 82.3%
- Indian Music: **72.2%**
- FMA Small: 41.1%



Cross-Validation Stability

Key Message: Consistent fold performance – narrow variance band.

- U-Net 5-fold CV on GTZAN (track-level splits)
- Mean: **90.44%** Std: **0.68 p.p.**
- Max/Min: 91.44% / 89.44% (range 2.0 p.p.)
- Supports reliability of reported single-split test score.



Conclusion

Conclusion & Key Takeaways

Key Message: We provide a rigorous benchmark and validate a strong architecture.

- Leak-free methodology is **essential** for trustworthy MGC results.
- U-Net encoder delivers 90.4% (CV) / 82.3% (test) on GTZAN.
- Transfer learning unlocks high performance in new domains (Tabla 96.5%).
- Provides a transparent, reproducible baseline for future work.

Thank You

Questions?

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GitHub: <https://github.com/alepot55/MGC-GTZAN>